

Yousef Kart

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SKILLS

- **Languages:** C++17/20, Python, Java, TypeScript, SQL
- **Graphics:** OpenGL, Qt 6, Three.js, ImGui, OpenCV
- **Tools:** CMake, FastAPI, Git, Docker, Supabase, PostgreSQL

EXPERIENCE

Full Stack Software Developer – Red Piston – Windsor, ON Apr 2026 – Present

- Developed a multi-object 3D rendering feature for an augmented-reality platform used by 1,000+ users
- Built and shipped production web and mobile software across Next.js, React Native, Supabase, and Strapi
- Rebuilt a legacy Android application as a cross-platform React Native application for iOS, Android, and web, improving load performance and enabling real-time client monitoring

Software Engineer – Scelta – Windsor, ON Feb 2025 – Sept 2025

- Developed a real-time 3D geospatial visualization pipeline for rendering development models on web-based maps
- Reduced REDMap's initial load time by 97% through database query optimization and API endpoint restructuring
- Led development of a shared UI component library used across multiple production applications, eliminating hundreds of duplicate files
- Designed a production-ready Next.js application architecture that reduced new-project build time by more than 50%

3D Modeler & Designer – Freelance June 2021 – Feb 2025

- Delivered hundreds of optimized 3D assets for 40+ global clients, optimizing geometry, topology, and texture workflows for real-time performance
- Produced architectural visualizations for developments valued at \$100M+, reducing asset turnaround times by 66%

PROJECTS

Horizon 1.0 – Python, FastAPI, PostgreSQL, LLMs, Finnhub, yfinance

- Engineered an event-driven market intelligence pipeline that ingests financial news, classifies market events using LLMs, collects market data, and ranks potential trading opportunities
- Designed a scoring system combining event quality, relative volume, abnormal price movement, liquidity, market capitalization, and historical behaviour

Monte Carlo Simulation Engine – C++20, OpenGL, GLFW, ImGui, CMake

- Developed a C++ stochastic simulation engine implementing Geometric Brownian Motion for Monte Carlo price modelling
- Built real-time OpenGL visualization and interactive ImGui controls for exploring simulated price paths

Black-Scholes Model Visualizer – C++17, Qt 6, CMake

- Implemented a Black-Scholes options-pricing engine with analytical Delta, Gamma, Vega, Theta, and Rho calculations
- Built interactive Qt visualizations for real-time analysis of option values and Greek sensitivities across changing market parameters

Flag Frenzy – Lua, Roblox Studio, Blender

- Built and launched a multiplayer capture-the-flag shooter with custom gameplay systems and optimized 3D content
- Scaled to 2.5M monthly visits before the project was acquired for \$50K USD

EDUCATION

University of Windsor

Sept 2021 – May 2025

Bachelor of Computer Science (Honours) & Minor in Mathematics

GPA: 3.9/4.0 | Distinction | 4x Dean's Honour Roll | LEAD Medallion